Solution Exam Preparation

Task 1

Which configurative features are visible on the three-side view and the illustration of each aircraft? What are the corresponding derived properties for the aircraft? List at least 6 for the Piaggio P180 moderate speed, laminar flow executive transport aircraft.

Configurative feature	**Derived property**
Three-surface configuration	Improves stall safety as forward wing it stalls before the main wing, producing an automatic nosedown effect Main wing is reduced in size resulting in drag and weight reduction
Mid-wing configuration	Less interference drag than other wing position
High aspect ratio wing	Less induced drag
Higher incidence angle (wrt the fuselage) of canard than of main wing	Canard stalls before wing and thus a nose-down pitching moment is generated preventing the pilot from stalling the wing
Flaps on wing and canard	Low-speed properties; by wing flap deflection generated pitching moment is balanced out with canard flap deflection
Unswept wings	Beneficial for laminar flow (no crossflow transition)
Higher sweep in horizontal tail than in main wing	Horizontal tail stalls after main wing; critical Mach number of horizontal tail is higher to avoid loss of elevator effectiveness
Fuselage shape similar to airfoil	Beneficial for laminar flow (smoothly changing shape)
Turboprop engines	Suitable for low speeds, better SFC than turbofan at low speed flight
T-Tail configuration	Horizontal tail away from engine exhaust, end-plate effect on vertical tail, less downwash from wings
Short landing gear	Autonomous boarding at any airport, lighter than longer landing gear
Pusher engine configuration	Less airflow disturbance on wing → beneficial for laminar flow; propeller and engine exhaust behind cabin → lower cabin noise

Task 2

Determine the lift coefficient slope of the horizontal stabilizer at $M = 0.5$ *on aircraft level. Due to the vertical position, the local dynamic pressure is reduced by only 5% compared to the main wings and the effective angle of attack is reduced by 10%. If more than one method is applicable, use the most precise one.*

Lift coefficient slope (Polhamus method):

$$C_{L\alpha,HT}{}' = \frac{2\pi \cdot \mathrm{AR_{HT}}}{2 + \sqrt{\frac{\mathrm{AR_{HT}}^2 \cdot \beta^2}{\mathrm{K}^2} \cdot \left(1 + \frac{\tan^2 \Lambda_{50\%,\mathrm{HT}}}{\beta^2}\right) + 4}}$$

With:

- $\beta = \sqrt{1 - M^2} = \sqrt{1 - 0.5^2} = 0.8660$
- $K = 1 + 0.75 \cdot \left(\frac{t}{c}\right)_{HT} = 1 + 0.75 \cdot (10.4\%) = 1.078$
- $AR_{HT} = 4.57$
- $\tan(\Lambda_{50\%,HT}) = \tan(\Lambda_{25\%,HT}) - \frac{4}{AR_{HT}} \cdot \left[\frac{50-25}{100} \cdot \frac{1-\lambda_{HT}}{1+\lambda_{HT}}\right] =$

$$= \tan(0.5145\, rad) - \frac{4}{4.57} \cdot \left[\frac{50 - 25}{100} \cdot \frac{1 - 0.58}{1 + 0.58}\right] = 0.5071$$

Therefore:

$$C_{L\alpha,HT}{}' = \frac{2\pi \cdot 4.57}{2 + \sqrt{\frac{4.57^2 \cdot 0.8660^2}{1.078^2} \cdot \left(1 + \frac{0.5071^2}{0.8660^2}\right) + 4}} = 4.285\ /rad$$

Conversion to aircraft level:

$$C_{L\alpha,HT} = C'_{L\alpha,HT} \cdot \frac{S_{HT}}{S_{ref}} \cdot \left({q_{HT}}/{q} \cdot {\partial\alpha_{HT}}/{\partial\alpha}\right)$$

$$= 4.285 \cdot \frac{3.83}{16} \cdot (0.95 \cdot 0.90) = 0.8770\ /rad$$

Task 3

Calculate the parasite drag area of the main wing during subsonic cruise flight at $h = 12000m$ *and* $M = 0.6$*. Assume laminar conditions prevail on 45% of the (camouflaged aluminum) wing chord and weight the equations for the coefficient of friction accordingly.*

Form factor:

Location of maximum airfoil thickness:

$$(x/c)_m = 40\% \implies \Lambda_m = \Lambda_{40\%}$$

$$\tan(\Lambda_{40\%}) = \tan(\Lambda_{25\%,W}) - \frac{4}{AR_W} \cdot \left[\frac{40-25}{100} \cdot \frac{1-\lambda_W}{1+\lambda_W}\right]$$

$$\Leftrightarrow \Lambda_{40\%} = \arctan\left[\tan(0.0194\,rad) - \frac{4}{11.96} \cdot \left[\frac{40-25}{100} \cdot \frac{1-0.371}{1+0.371}\right]\right]$$

$$= -0.0036rad \approx 0°$$

Form factor:

$$F_W = \left[1 + \frac{0.6}{(x/c)_m} \cdot \left(\frac{t}{c}\right)_W + 100 \cdot \left(\frac{t}{c}\right)_W^4\right] \cdot [1.34 \cdot M^{0.18} \cos(\Lambda_{40\%})^{0.28}]$$

$$\implies F_W = \left[1 + \frac{0.6}{0.4} \cdot (14.5\%)_W + 100 \cdot (14.5\%)_W^4\right] \cdot [1.34 \cdot 0.6^{0.18} \cos(0)^{0.28}] = 1.542$$

Skin friction coefficient:

Reference length:

$$c_{MAC} = 1.284\,m$$

Flight speed:

$$V = M \cdot a = 0.6 \cdot 295 = 177.0\,m/s$$

Reynolds number:

$$Re_W = \frac{\rho \cdot V \cdot c_{MAC}}{\mu} = \frac{0.311 \cdot 177 \cdot 1.284}{14.32 \cdot 10^{-6}} = 4.936 \cdot 10^6$$

Cut-off Reynolds number:

$$Re_{CO,\mathrm{W}} = 38 \cdot \left(\frac{c_{MAC}}{k}\right)^{1.053} = 38 \cdot \left(\frac{1.284}{10 \cdot 10^{-6}}\right)^{1.053} = 9.101 \cdot 10^6$$

Proceed with the smallest of Re_W and $Re_{CO,\mathrm{W}}$. In this case: Re_W

Skin friction coefficient:

- Laminar:

$$c_{f,W,lam} = \frac{1.328}{\sqrt{Re_W}} \text{ (Blasius equation)}$$

$$\implies c_{f,W,lam} = \frac{1.328}{\sqrt{4.936 \cdot 10^6}} = 0.0005978$$

- Turbulent:

$$c_{f,W,turb} = \frac{0.455}{(\log_{10} Re_\mathrm{W})^{2.58}} \cdot \left(1 + 0.144 \cdot {M_\infty}^2\right)^{-0.65} \text{ (Schlichting equation)}$$

$$\implies c_{f,W,turb} = \frac{0.455}{(\log_{10} 4.936 \cdot 10^6)^{2.58}} \cdot (1 + 0.144 \cdot 0.6^2)^{-0.65} = 0.003263$$

- Total (weighted average):

$$c_{f,W} = 0.45 \cdot c_{f,W,lam} + 0.55 \cdot c_{f,W,turb}$$
$$= 0.45 \cdot 0.0005978 + 0.55 \cdot 0.003263 = 0.002064$$

Wetted surface area:

Plate analogy area:

$$S_{wet,W} = 2 \cdot S_{ref} = 2 \cdot 16 = 32.00\ m^2$$

Wing parasite drag area:

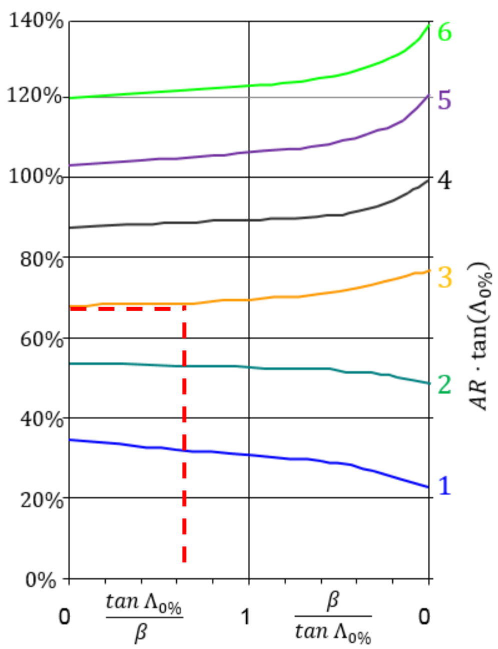
$$f_W = c_{f,W} \cdot F_W \cdot S_{wet,W} = 0.002064 \cdot 1.542 \cdot 32 = 0.1018\ \mathrm{m}^2$$

Task 4

Locate the absolute location of the aerodynamic center of the horizontal tail during approach ($M = 0.3$), measured from the nose of the aircraft.

Taper ratio $\lambda_{HT} = 0.58 \approx 0.5 \rightarrow$ choose figure accordingly

$$tan(\Lambda_{0\%,HT}) = tan(\Lambda_{25\%,HT}) - \frac{4}{AR_{HT}} \cdot \left[\frac{0-25}{100} \cdot \frac{1-\lambda_{HT}}{1+\lambda_{HT}}\right]$$
$$\Rightarrow tan(\Lambda_{0\%,HT}) = tan(0.5145\ rad) - \frac{4}{4.57} \cdot \left[\frac{0-25}{100} \cdot \frac{1-0.58}{1+0.58}\right] = 0.6234$$



x-axis: *with* $\beta = \sqrt{1-\mathrm{M}^2} = \sqrt{1-0.3^2} = 0.9539$

$$\frac{tan\ \Lambda_{0\%,HT}}{\beta} = \frac{0.6234}{0.9539} = 0.6535$$
$$\frac{\beta}{tan\ \Lambda_{0\%,HT}} == \frac{0.9539}{0.6234} = 1.530$$

$\rightarrow$ Take value that's between 0 and 1

curve:

$AR_{HT} \cdot \tan(\Lambda_{0\%,HT}) = 4.57 \cdot 0.6234 = 2.849 \approx 3$

y-axis:

$$n_{AC,HT} \cong 70\%$$
$$\mathrm{x_{AC,HT}} = \mathrm{x_{0,HT}} + \mathrm{x_{ap,HT}}$$
$$= \mathrm{x_{0,HT}} + (\mathrm{n_{AC,HT}} \cdot \mathrm{c_{root,HT}})$$
$$\Rightarrow \mathrm{x_{AC,HT}} = 12.07 + (70\% \cdot 1.10) = 12.84\mathrm{m}$$

Task 5

Estimate the mass of the fuselage in kilograms. The design dive speed is $400\ kn$.
Equation to be used:

$$m_{fus} = 0.021 \cdot C_{fus} \cdot \sqrt{V_D \cdot \frac{l_{HT}}{w_F + h_{fus}}} \cdot S_{fus}^{1.2}$$

With:

- $C_{fus} = 1.08 \cdot 1.07$ (pressurized cabin and main landing gear attached to fuselage, engines on wing)
- $d_{HT} \approx x_{0,HT} - x_{0,W} = 39.60 - 23.92 = 15.68\ ft$
- $l_{fus} = 47.24\ ft$
- $w_{fus} = 6.069\ ft$
- $h_{fus} = 5.741\ ft$
- $S_{fus} = l_{fus} \cdot \pi \cdot \frac{w_{fus} + h_{fus}}{2} = 47.24 \cdot \pi \cdot \frac{6.069+5.741}{2} = 876.4\ ft^2$
- $V_D = 400\ kn$

Therefore:

$$m_{fus} = 0.021 \cdot 1.08 \cdot 1.07 \cdot \sqrt{400 \cdot \frac{15.68}{6.069 + 5.741}} \cdot 876.4^{1.2} = 1900\ lbm = 861.8\ kg$$

Task 6

An upgraded version of the aircraft shall use two turbofan engines in order to reach a nominal thrust to weight ratio of $T_0/w_{TO} = 0.4$. *With the help of the sizing equations, estimate the length of the new nacelles. Cruise Mach number* $M = 0.6$. *Assume that the BPR is lower than 6.*

$$\frac{T_0}{w_{TO}} = 0.4 \Longrightarrow T_0 = 0.4 \cdot w_{TO} = 0.4 \cdot m_{TO} \cdot g = 0.4 \cdot\ 5239 \cdot 9.81 = 20560\ N$$

From three view: $n_{eng} = 2$. Therefore:

$$T_{unit} = \frac{T_0}{n_{eng}} = \frac{20560}{2} = 10280\ N$$

Nacelle length calculation:

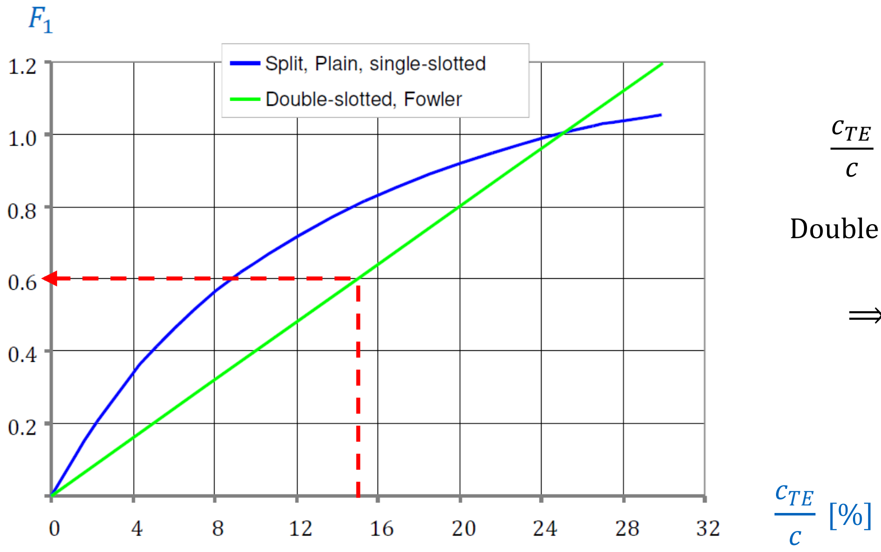
$$l_{nac} = 0.49 \cdot \left(\frac{T_{unit}}{1000}\right)^{0.4} \cdot \mathrm{M}^{0.2}$$
$$= 0.49 \cdot \left(\frac{10280}{1000}\right)^{0.4} \cdot 0.6^{0.2} = 1.124\ \mathrm{m}$$

Task 7

Estimate the maximum lift coefficient increment when the (average) double-slotted trailing edge flaps are deflected by $\delta_{TE} = 20°$. Assume that 80% of the wing area is influenced by the flaps, which have a chord of $c_{TE} = 0.2\ m$.

1) Airfoil:

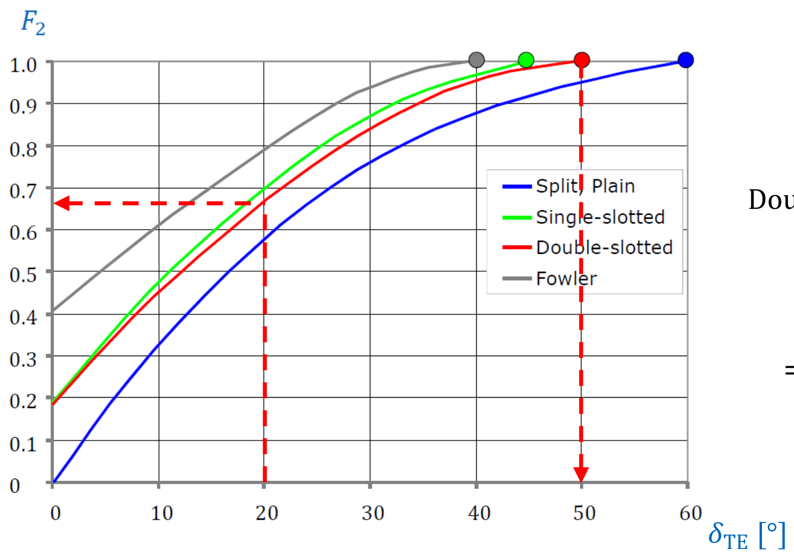
$$\Delta c_{l,\max(TE)} = F_1 \cdot F_2 \cdot F_3 \cdot \Delta c_{l,max,base}$$



$$\frac{c_{TE}}{c} = \frac{0.2}{1.284} = 15.58\ \%$$

Double slotted → green curve

$$\Longrightarrow F_1 = 0.6$$

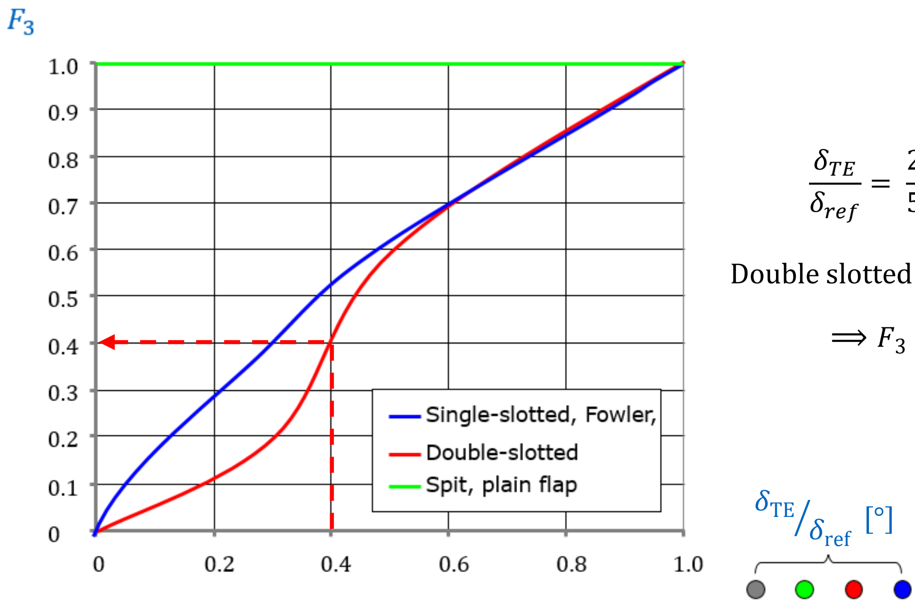


$$\delta_{TE} = 20°$$

Double slotted → red curve

$$\Longrightarrow F_2 = 0.65$$

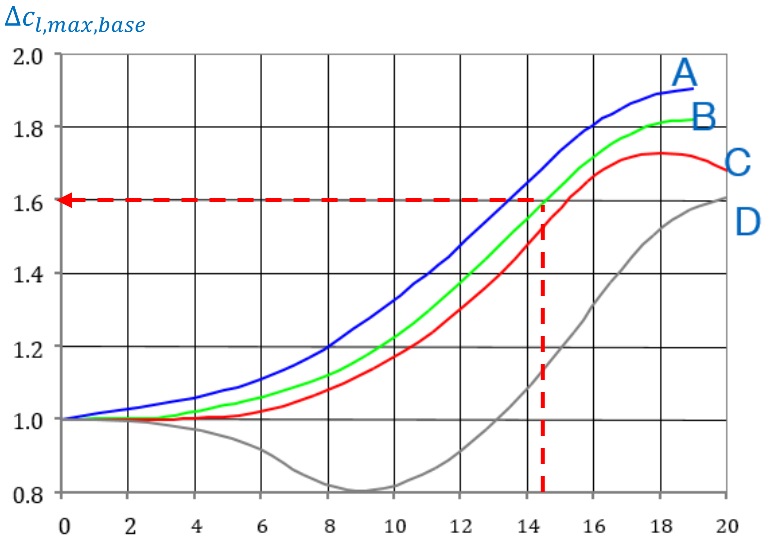
$$\Longrightarrow \delta_{ref} = 50° \ @$$



$$\frac{\delta_{TE}}{\delta_{ref}} = \frac{20°}{50°} = 0.4$$

Double slotted $\rightarrow$ red curve

$$\Longrightarrow F_3 = 0.4$$



$\frac{t}{c}$ [%]

$$\frac{t}{c} = 14.5\ \%$$

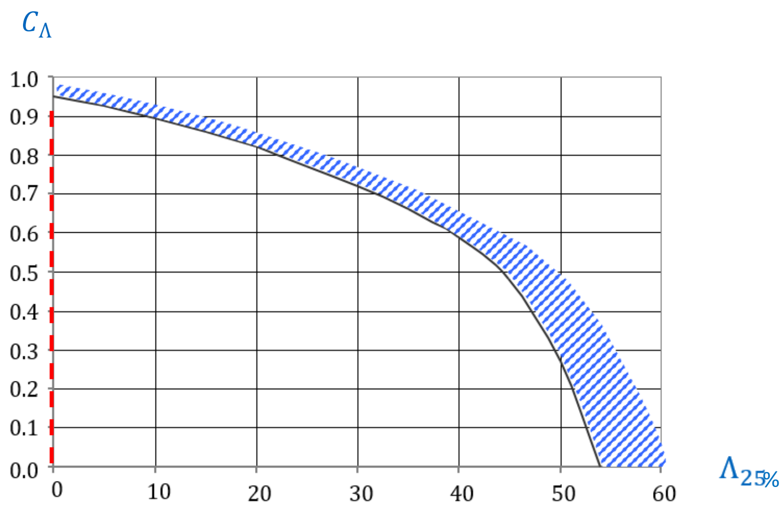
Double slotted $\rightarrow$ curve B (see legend in problem formulation)

$$\Longrightarrow \Delta c_{l,max,base} = 1.6$$

$$\Longrightarrow \Delta c_{l,\max(TE)} = F_1 \cdot F_2 \cdot F_3 \cdot \Delta c_{l,max,base} = 0.6 \cdot 0.65 \cdot 0.4 \cdot 1.6 = 0.2496$$

2) Wing:

$$\Delta C_{L,max(TE)} = \Delta c_{l,max(TE)} \cdot \frac{S_{W,TE}}{S_{ref}} \cdot C_\Lambda$$



$$\Lambda_{25\%} = 1.1°$$

$$\Longrightarrow C_\Lambda = 0.95$$

$$\Longrightarrow \Delta C_{L,\max(TE)} = \Delta c_{l,\max(TE)} \cdot \frac{S_{W,TE}}{S_{ref}} \cdot C_\Lambda = 0.2496 \cdot 80\% \cdot 0.95 = 0.1897$$

Task 8

Draw the limit curve for a take-off distance of $s_{acc} = 650\ m$ *at an altitude of* $2000\ m$ *above mean sea level into a design chart (limit the number of supporting points of the resulting curve to three) and indicate the location of the design region.* $C_{L,max} = 2.2$, $C_{D0|tot} = 0.016$, $C = 1.15$, $C_{L,TO} = 1.66$, $\mu_f = 0.02$.

Equation to be used (derivation see lecture slides):

$$\frac{T_0}{w_{TO}} \geq \frac{C^2}{C_{L,max,TO} \cdot s_{\text{acc}} \cdot \rho \cdot g} \cdot \frac{w_{TO}}{S} + \frac{C^2}{2 \cdot C_{L,max,TO}} \cdot \left(C_{D0|tot} + \frac{C_{L,TO}^2}{3 \cdot \pi \cdot AR \cdot e_{TO}} - \mu_f \cdot \mathrm{C}_{L,TO} \right) + \mu_f$$

With:

- $C = 1.15$
- $C_{L,max,TO} = 2.2$
- $s_{acc} = 650\ m$
- $\rho = 1.006\ kg/m^3$ (from standard atmosphere)
- $g = 9.81°m/s^2$
- $C_{D0|tot} = 0.016$
- $C_{L,TO} = 1.66$
- $AR = 11.96$
- $e_{TO} = 0.80$
- $\mu_f = 0.02$

Therefore:

$$\frac{T_0}{w_{TO}} = \frac{1.15^2}{2.2 \cdot 650 \cdot 1.006 \cdot 9.81} \cdot \frac{w_{TO}}{S} + \frac{1.15^2}{2 \cdot 2.2} \cdot \left(0.016 + \frac{1.66^2}{3 \cdot \pi \cdot 11.96 \cdot 0.80} - 0.02 \cdot 1.66\right) + 0.02$$

$$= 9.371 \cdot 10^{-5} \cdot \frac{w_{TO}}{S} + 0.0240$$

Plot: